Module 15: Population and Systems Ecology - Key Points

Learning Objectives

By the end of this module, you should be able to:

1. Compare exponential and logistic population growth.
2. Explain carrying capacity and limiting factors.
3. Distinguish density-dependent and density-independent regulation.
4. Use trophic levels and the 10 percent rule to explain energy limits.
5. Contrast energy flow with nutrient cycling.

Topics

- Population growth models
- Limits and carrying capacity
- Community interactions
- Energy flow through trophic levels
- Matter cycling and systems feedback

Key Terms to Know

- **Population** - Members of one species in an area.
- **Exponential growth** - Growth pattern with increasing rate under abundant resources.
- **Logistic growth** - Growth that slows as population approaches carrying capacity.
- **Carrying capacity** - Approximate environment-supported population size.
- **Density-dependent factor** - Limiting factor whose impact changes with crowding.
- **Density-independent factor** - Limiting factor not directly tied to density.
- **Trophic level** - Feeding position in an energy pathway.
- **Decomposer** - Organism that breaks down dead material and recycles nutrients.

Core Contents

1. Births and immigration add individuals; deaths and emigration remove them. Compare counts over the same area and time interval.
2. Exponential per-capita growth is constant; logistic per-capita growth slows near carrying capacity K. Total logistic growth peaks near K/2.
3. Density-dependent effects vary with crowding. A freeze with equal proportional impact at low and high density is density-independent.
4. Assume 10 percent transfer per trophic step for calculations; real efficiencies vary. From 10000 kJ in producers, primary consumers receive 1000 kJ.
5. Matter cycles; usable energy must enter and dissipates as heat. Decomposers return nutrients. Competition and resource feedback connect populations.

Study Tips

1. Complete Lab 16 as a prediction, evidence check, and revision exercise. No internet or special equipment is needed.
2. Answer the learning questions as homework practice; show reasoning and units when a question supplies numbers.
3. Use Practice Test 03, then check its self-contained key and explain any correction in your own words.
4. Connect each answer to a mechanism and evidence, rather than a memorized definition alone.

Connected Lab

- `lab-16_population-systems-ecology.md`

Generated Visuals

- Module 15: Population and Systems Ecology Evidence Map
- Module 15: Population and Systems Ecology Reasoning Sequence
- Module 15: Population and Systems Ecology Retrieval and Lab Check